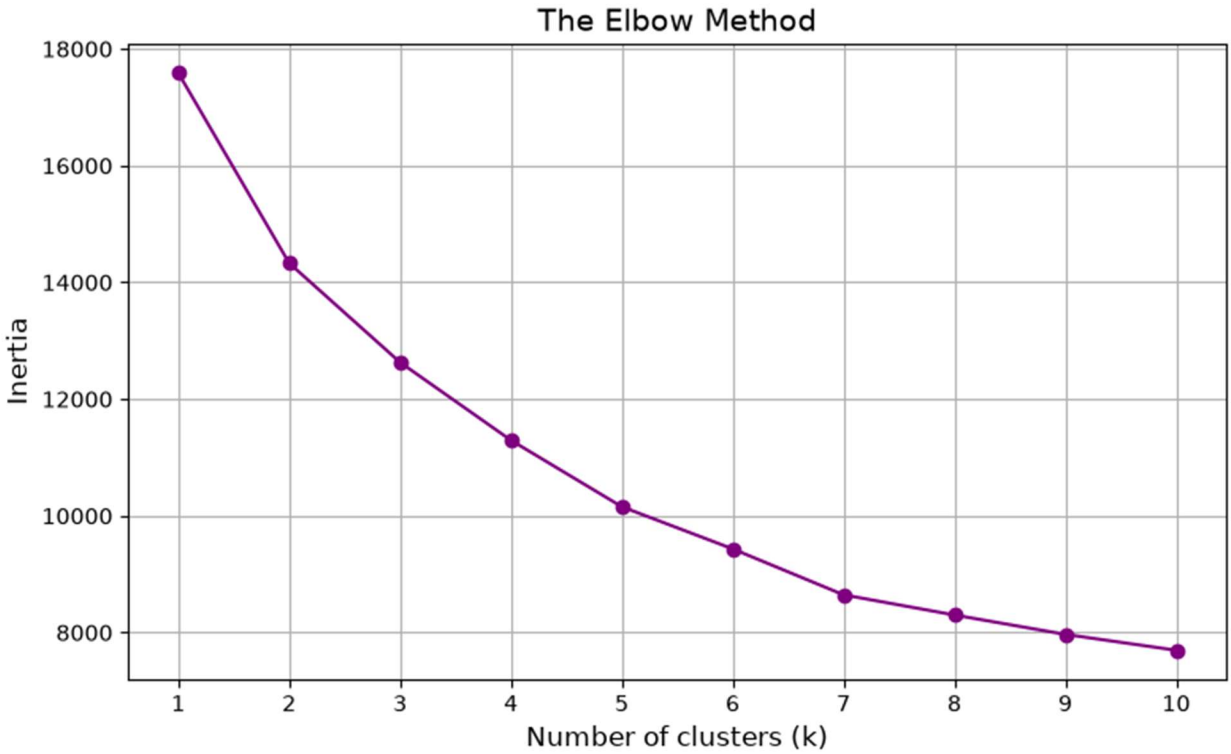
ECE 607

Homework #5

Dr. Jiang Li

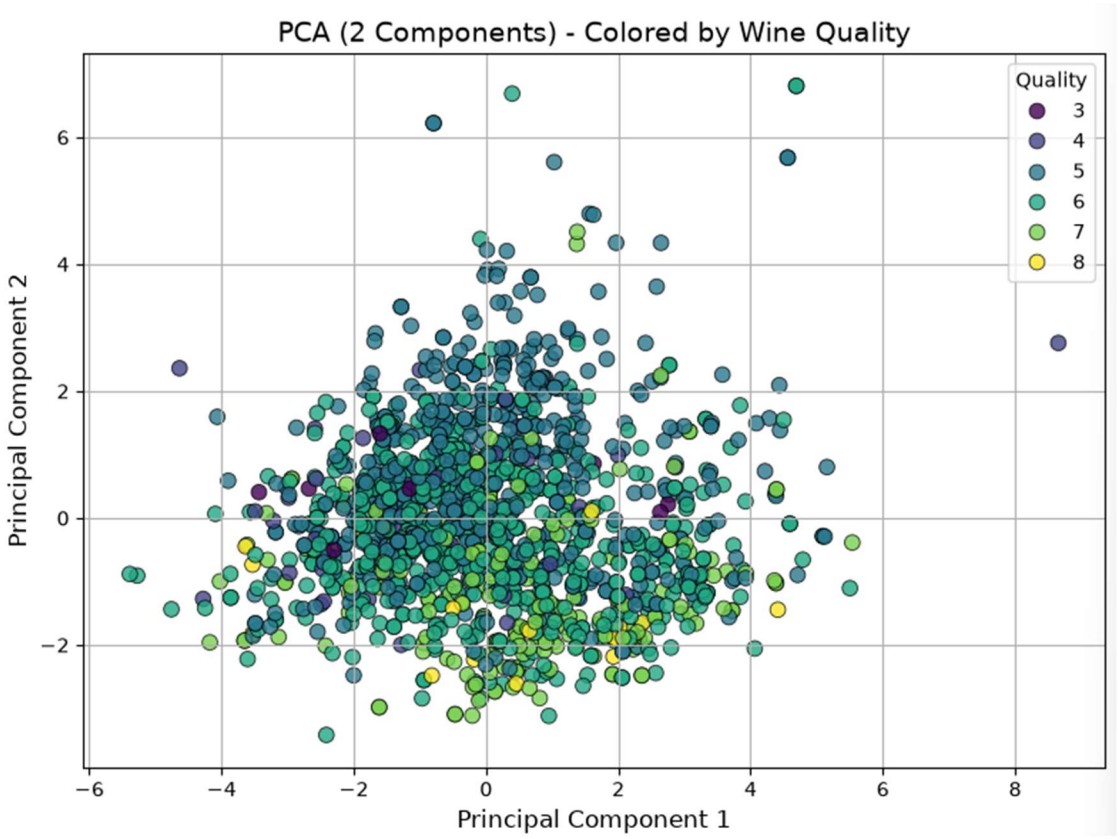
Austin Alexander Broome

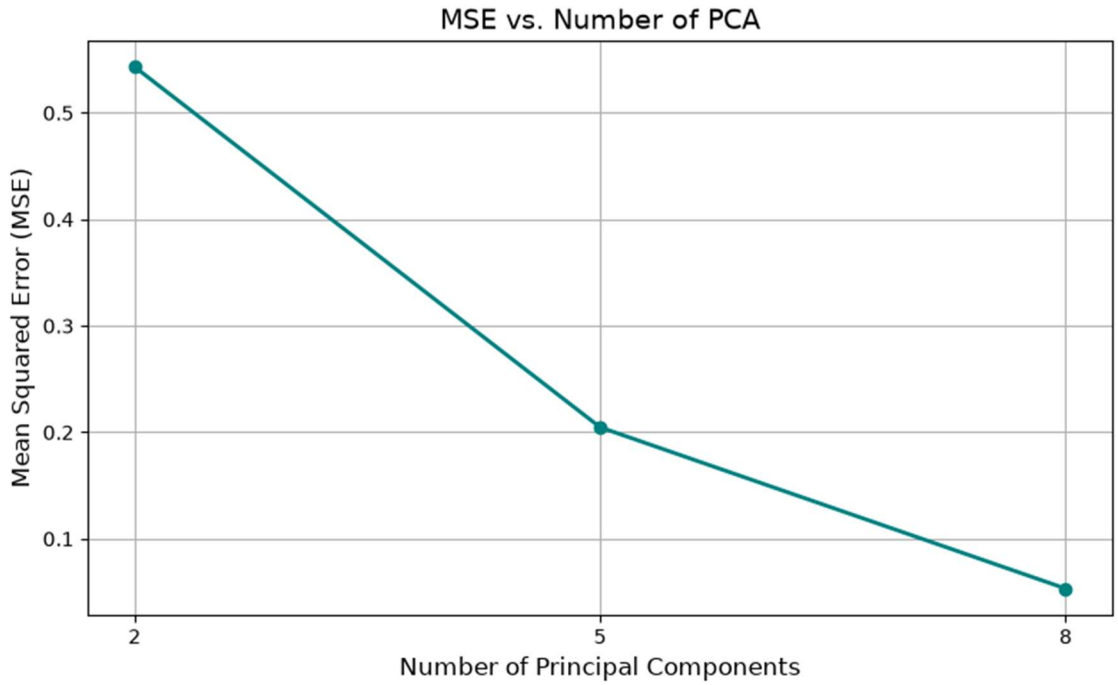
1)



Based on the elbow method I determine the optimal k to be 6.

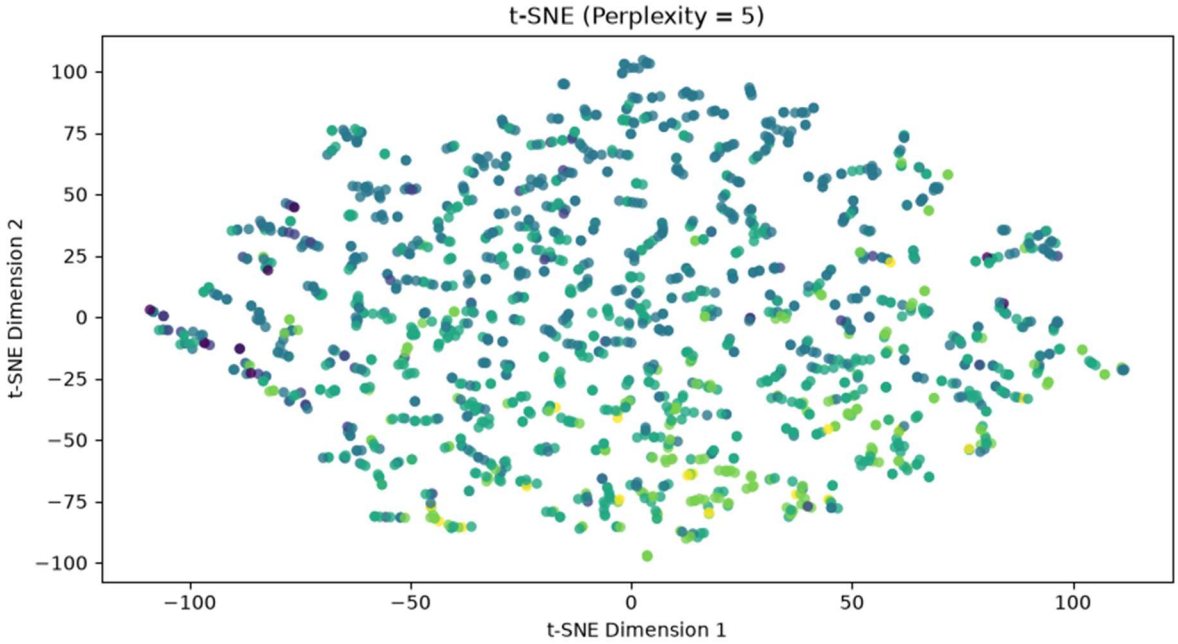
2)

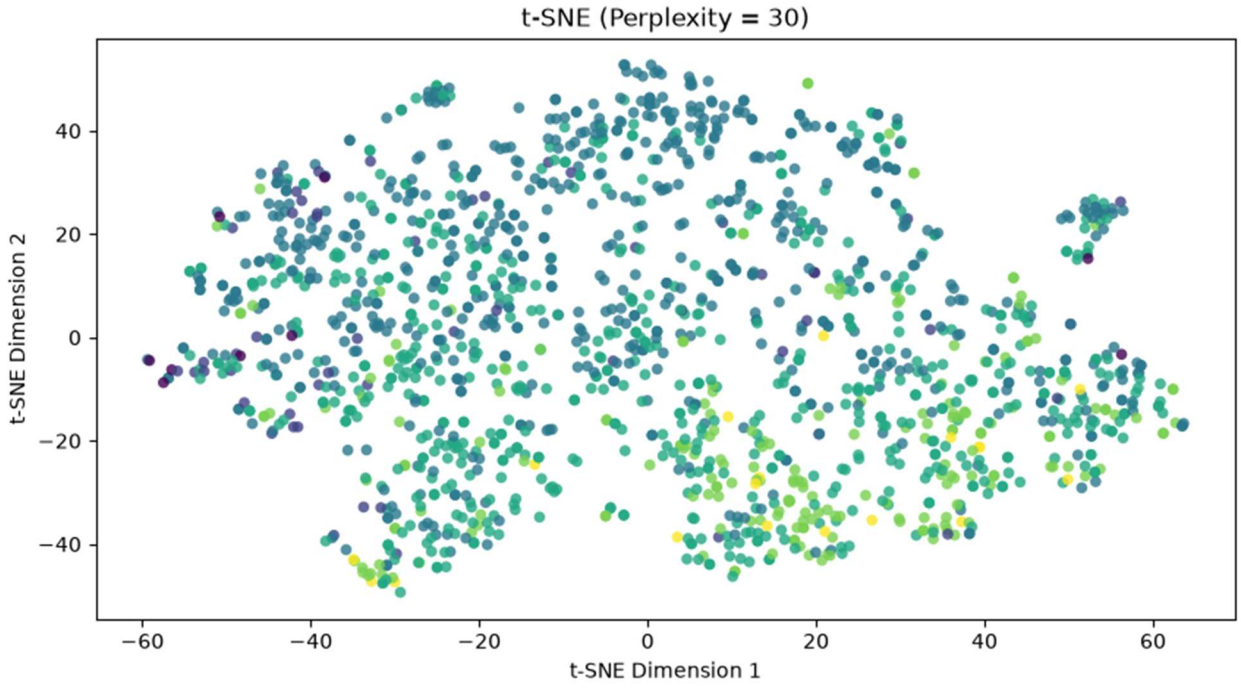

PCA (2 Components) - Colored by Wine Quality
Quality
3
4
5
6
7
8
Principal Component 2
Principal Component 1
6
4
2
0
−2
−6
−4
−2
0
2
4
6
8

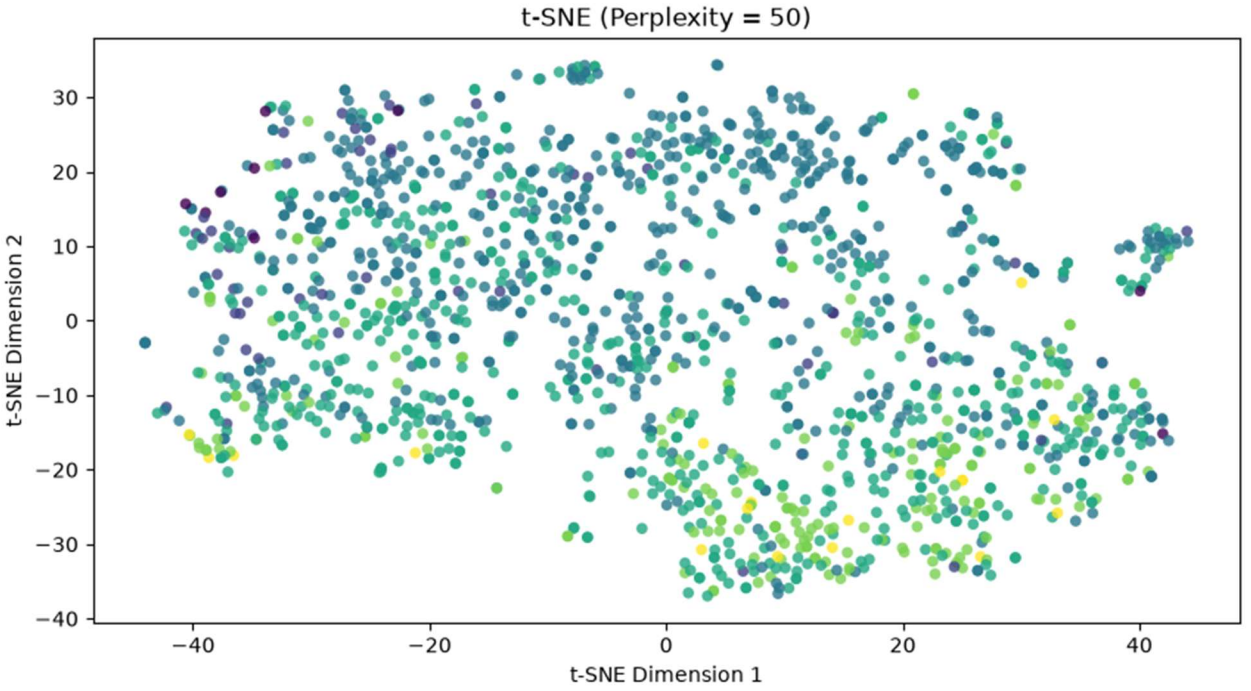

MSE vs. Number of PCA
Mean Squared Error (MSE)
Number of Principal Components
0.5
0.4
0.3
0.2
0.1
2
5
8

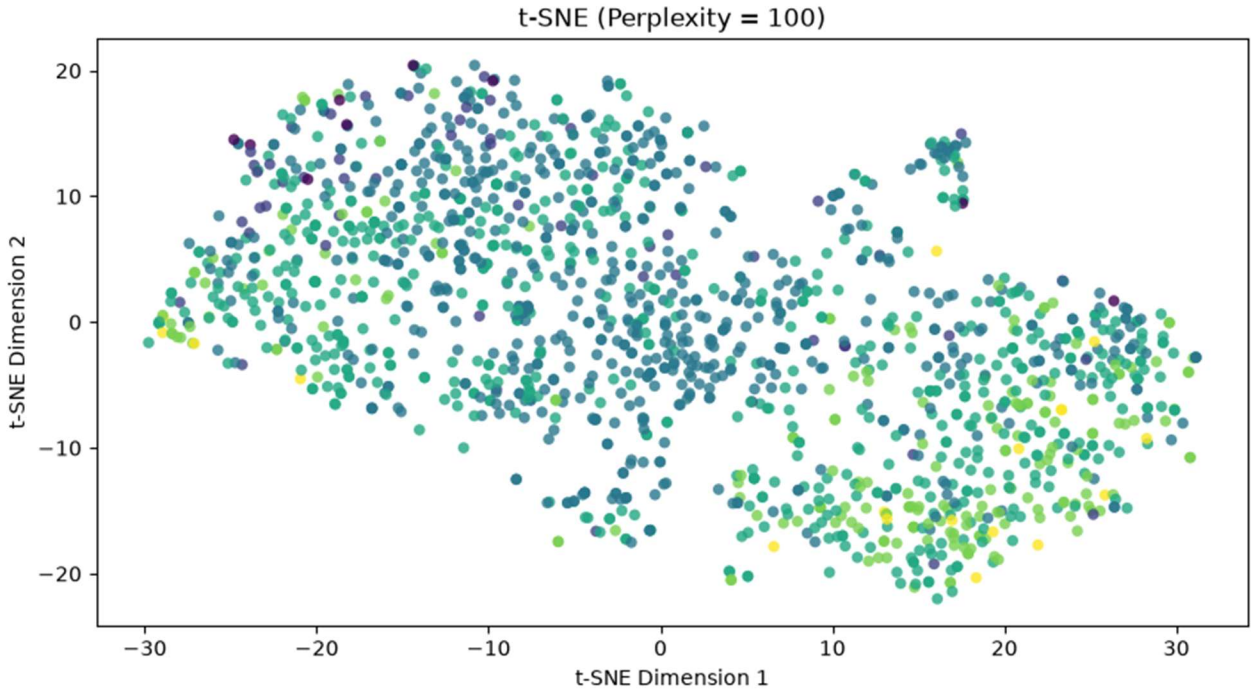
Principal Component Analysis	Mean Square Error
2	0.5432
5	0.2047
8	0.0532

3)









Lower values of perplexity seem to result in clusters that are local and apart from other clusters. As the value of perplexity is increased the clusters seem to be closer and smooth out.